# *Ecology* Data Paper Author Guidelines

**Note:** These guidelines should be used for Data Papers that were submitted after Spring 2026 and prepared in the updated Data Paper format. Data Papers submitted through Spring 2026 follow the format in earlier guidelines. Complete guidelines are available on the *Ecology* Data Papers hub.

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## What are Data Papers?

Data Papers are a unique type of manuscript published in *Ecology*, designed to enrich the global ecological and scientific community through the sharing of data to be reused, reanalyzed, and remixed for advancing knowledge beyond its original purpose. This definition emphasizes reusability as the key characteristic over large or expansive data collection efforts. Submissions must clearly identify ecological application(s), impact, and scope. Secondary data (i.e., data compiled from other published sources) is encouraged if it represents a significant, innovative, or creative contribution. Submissions should include descriptive documentation (scientific context) and technical documentation (metadata) and follow the data and metadata standards described in these guidelines. Data Papers are subject to full peer review. The review process will evaluate ecological significance and overall quality first, with further technical review to ensure a high standard of usability, especially with respect to associated metadata.

## Open Research Policy Requirements

In accordance with our Open Research Policy, data being presented in a Data Paper must be:

1) Uploaded to a suitable data repository when Data Papers are **submitted** (with data either fully published or in private-for-peer-review status for this evaluation)
2) Published and publicly available in a suitable data repository for **accepted** Data Papers
3) Data repository selection for Data Papers
    a. The following repositories are pre-approved.

| **Generalist repositories** | **Specialist repositories** |
|---|---|
| Dryad | Genbank |
| Zenodo | NCBI |
| Figshare | Movebank |
| OSF | TERN |
| EDI | GBIF |
| USGS ScienceBase | vectorbase |
| | wormbase |

    b. Other options for repositories may be approved for Data Papers. It is expected that repositories meet specific criteria:
        i. persistent identifier (DOI preferred)
        ii. commitment to long-term storage of the data (20+ years)
        iii. authors retain copyright and choose an open license (https://creativecommons.org/share-your-work/)
        iv. ORCiD integration

v. data access is free of charge
vi. public access preferred; controlled access is allowable under some circumstances (e.g., endangered species location; confidentiality; human subject data)

Additionally, in accordance with our Open Research Policy, any code used to process and/or summarize data and generate the tables and figures in the Data Paper must be provided in a suitable repository.

## ESA Publication Policies

Authors, Reviewers, and Editors are required to follow these policies established by the Ecological Society of America (ESA). These include ESA's Code of Ethics.

## Licensing Options and Fees

1) Standard journal publication licensing options and publishing fees apply to the material typeset for publication in *Ecology*.

   Licensing information: https://www.esa.org/publications/our-policies/licensing/

   Fee information: https://www.esa.org/publications/our-policies/publication-fee/

2) Authors can deposit their data and code associated with *Ecology* Data Papers in Dryad at no cost (up to 10 GB; materials must be curated by Dryad after January 1, 2026 to qualify for fee coverage under ESA sponsorship). We encourage you to explore Dryad's services and consider this option for data and code archiving purposes.
3) Authors using other external repositories are responsible for any fees charged by that repository.

## Organizing Your Data Paper Files

A Data Paper is composed of four or five elements: (1) a manuscript file; (2) a metadata file; (3) a README file; (4) the data files; (5) code files (when applicable). **The manuscript file is the only component that is submitted via ScholarOne for typesetting and publication in *Ecology*. All other files are uploaded to an external data repository, and the link to that data repository must be provided in the manuscript, as described herein.**

## Manuscript

- The manuscript file is submitted via ScholarOne as the “Main Document” and is the only part of the Data Paper that will be typeset for publication in *Ecology*.
- The manuscript should have the following sections: Title page; Abstract (500 words maximum); Keywords; Overview; Data Availability Statement; Data Description; Summary of Methods; Limitations and Challenges; Scientific Opportunities; Tables; Figures; References.
- The manuscript can be a maximum of 2000 words, excluding the title page, abstract, tables, figure legends, and the references section.
- This template should be followed to prepare the manuscript: template_manuscript_data_paper
- See the “Manuscript formatting guidelines” section for complete instructions.

## Metadata

- The Metadata is **not** submitted via ScholarOne and is **not** published in *Ecology*. It is published on the data repository with the data.
- The Metadata consists of a narrative document describing the data.
- The Metadata document should be named “Metadata”, with the appropriate file extension.
- This template should be followed to prepare the Metadata: template_metadata_data_paper
- See the “Metadata Formatting” section for more information.

## README

- The README is **not** submitted via ScholarOne and is **not** published in *Ecology*. It is published on the data repository with the data.
- The README is a technical summary of the data designed to be a standard, machine-readable, interoperable list of basic data descriptors.
- Note that this is separate from the Metadata file.
- The README document should be named "README", with the appropriate file extension, such as .md.
- This template should be followed to prepare the README file: template_README_data_paper
- See the "README Formatting" section for more information.

### Data and code files

- Data files are **not** submitted via ScholarOne and are **not** published in *Ecology*. These are published on the data repository.
- Data files should be consistently formatted in nonproprietary file formats (such as comma-separated values [.csv] or tab-delimited ASCII text [.txt]).
- Each file must be given a unique and descriptive file name.
- Data files must be available with the original submission to the journal. Data may be provided in private-for-peer-review status for that stage. For accepted Data Papers, data must be published and openly available in a data repository (with limited exceptions). See the "Open Research Policy requirements" section for a list of suitable repositories.
- Information from suggested repositories with a private sharing-for-peer-review option is provided below:
    - Generate a private-for-peer-review link in Dryad: https://datadryad.org/help/submission_steps/peer_review
    - Generate a private-for-peer-review link in Figshare: https://info.figshare.com/user-guide/how-to-use-private-links/
    - Generate a "secret link" in Zenodo: https://help.zenodo.org/docs/share/link-sharing/
- Code should be provided in accordance with ESA's Open Research policy. Novel code or computer software utilized to generate data packages is required for peer review. All methods or protocols utilized to generate data packages must be disclosed, and code for new methods or protocols is required for peer review. Existing methods or protocols must be cited (including references).

## *Ecology* Data Paper Author Checklist

This checklist outlines the submission requirements to guide this process.

## Manuscript Formatting Guidelines

The manuscript is the only part of a Data Paper that is typeset. It includes the Title page, Author and data compilers list, Abstract, Keywords, Overview, Data Availability Statement, Data Description, Summary of Methods, Limitations and Challenges, Scientific Opportunities, Tables, Figures, and References. The title page includes additional elements for publication standards. The manuscript is designated as the "Main Document" in the file list. The manuscript can be a maximum of 2000 words (excluding the title page, abstract, tables, figure captions, and the

references section). The manuscript should be provided in Word file format. This template should be followed to prepare the manuscript.

## Title Page

### Title

- Titles should be concise and informative. The maximum length is 120 characters (including spaces).
    - ScholarOne enforces this limit in the “Title” field of the ScholarOne online submission form.
    - If your title does not fit into this field, it must be shortened, and the manuscript should be changed to reflect the shortened title.
- Use sentence case, capitalizing only the first word and proper nouns. If the title consists of two parts separated by a colon, capitalize the first word following the colon.

### Authors and Data Compilers

- For each author, state the relevant address—usually the institutional affiliation of the author during the period when all or most of the data were collected.
    - Each author’s present address(es), if different from this, should appear at the bottom of the title page and be labeled as “Present Address for [Author Name].”
- Note the corresponding author on the title page.
    - Provide a single email address for the corresponding author to which questions regarding the dataset can be directed.
    - There can be only one corresponding author on a manuscript.
- Please take extra care with the order and spelling of author names and affiliations, especially if the author list is lengthy. To expedite the manuscript intake process, please closely check the following:
    - Author names listed in the manuscript and in the ScholarOne online submission form should match.
    - Author names must be provided in the same order between the manuscript and the ScholarOne online submission form.

### Keywords/phrases

- Please supply a minimum of 6 and a maximum of 12 keywords/phrases for indexing purposes.

- The list of keywords/phrases in the manuscript, metadata, README, and the ScholarOne online submission form should match.
- The list of keywords/phrases should be in alphabetical order in all locations.

### Acknowledgments

- Detail all funders that supported this work.
- In recognition of the critical role that field stations, marine labs (FSMLs) and related research institutions and facilities serve in ecology, we ask that authors who have conducted research at or with the support of FSMLs provide the names and locations of FSMLs in both the Methods and Acknowledgments sections of the manuscript. By asking authors to standardize how they report the support provided by FSMLs, readers will more easily and accurately understand where data were collected.
- List any additional acknowledgments.

### Contributor Roles

- Define the role of each author during this work using CRediT taxonomy.

### Conflict of Interest

- Include a brief statement indicating whether any Conflicts of Interest exist and specify any that do.

## Abstract

- The *Abstract* should be a concise summary of the Data Paper (maximum 500 words), highlighting the main points brought up in the manuscript text.
- The *Abstract* should provide a synthesis of each section contained in the main manuscript (Overview, etc.)

## Overview

- The *Overview* should identify the information gap that the data are filling and the motivation for publishing them.
- The *Overview* should relate the submission to existing data, research, or outstanding problems in the field.

## Data Availability Statement

- The *Data Availability Statement* provides the DOI of the repository where the data, code (when applicable), metadata, and README are stored, as well as the license and any other accessibility information that may apply.
- See our Open Research policy for additional information.

## Data Description

- The *Data Description* section should provide a concise description of the data (e.g., focal phenomena; type of data; variables measured; spatial, temporal, and taxonomic extent; number of tables and relationships between tables; number of rows and columns in each table and a brief description of their content). It should be strictly descriptive, with no statistical analysis or inferential statements.

## Summary of Methods

- The *Summary of Methods* section should provide a brief narrative description of how the data were collected and the experimental or sampling design. Details on the data collection protocol belong in the Metadata file.

## Limitations and Challenges

- The *Limitations and Challenges* section should highlight potential pitfalls that future users should be aware of when reusing the data. For instance, biases inherent to the data (and how they can be mitigated), missing information (and how it might impact inference), descriptions of measurement uncertainty and its quantification, or any other anomalies that users should be cautious about.

## Scientific Opportunities

- The *Scientific Opportunities* section should provide the authors' vision for how the data can be reused by others to answer new ecological questions. It needs to highlight the value of the Data Paper contribution by describing potential avenues for application of the data.

## Tables and Figures

- The manuscript can include a combined maximum of no more than four tables and figures.
- Tables and figures should be strictly descriptive (no results from statistical analyses).
- Please refer to *Ecology*'s author guidelines for general table and figure formatting details.

### References

- Include a complete References section for all works cited in the manuscript.
- Journal style follows a variation of Chicago Reference Style and the Author Year citation format. Detailed styling is applied by the publisher, but please ensure all elements for each reference are provided.

## Metadata Formatting Guidelines

Metadata should describe the content, context, quality, and structure of the data in a manner that facilitates use of the data by the research community. The Metadata document should be composed using our template document in plain text (".txt") or markdown (".md") format. Metadata fields should be completed where appropriate.

### General Guidelines for Metadata Sections

#### Identifiers

The essential identifying information for the dataset and its authors. These contents should directly match the README file.

#### Research Origin Descriptors

The motivation or purpose of your research should appear here.

**Study Design**

State the objectives of the data collection (e.g., questions; hypotheses; relevant background), any project titles, and relevant funding mechanisms.

**Subproject/Sampling Site Descriptors**

This section should accompany each relevant subproject or sampling site (i.e., for a dataset that may span multiple data collection sessions and/or geographical sites).

It should include the following:

- A description of the geography, geology, habitat, and climate, as relevant to the research project.
- A description of the research methods, experimental design, sampling procedures, instrumentation, and data entry/quality/assurance protocols.

#### Data Structure Descriptors

This section describes the structural organization of the dataset.

#### Data File Descriptors

Describe the file-level organization of the dataset (e.g., naming scheme; format; header information). If applicable, include a description of the schema for relationships between data files (e.g., unique key identifiers that link rows of data in one data file with rows of data in another data file).

#### Observation Information

Describe what the rows in each file encode. Multiple sections may be needed to describe the organization across different portions of the dataset.

#### Variable Information

Describe what the columns in each file encode, including the column name, type of data stored in the column (factor, numeric, date, etc.), description of the column, and range of values or levels. Multiple sections may be needed to describe the organization across different portions of the dataset.

### Supplemental Descriptors

#### Archiving

Describe any relevant history for the archival of the dataset.

#### History of Usage

Describe any notes related to historical usage of the dataset.

## README Formatting Guidelines

The README should contain a minimal technical summary of the data, necessary for basic reuse of the dataset. The README should be compiled using our template document in plain text (".txt") or markdown (".md") format. README fields should be completed where appropriate.

## Data Formatting Guidelines

Data should be logically and consistently formatted. The primary goal is to ensure that your files will be accessible and usable to every user, on every platform, for the foreseeable future. The following guidelines describe various components of your data files.

1. Tabular data (data that is readily organized as one or more spreadsheets) should follow the Tidy Data guidance (Wickham 2014; https://doi.org/10.18637/jss.v059.i10) that "each

variable is a column, each observation is a row, and each type of observational unit is a table". Exceptions are allowed on a case-by-case basis.

2. Data files should be in a nonproprietary format. Spreadsheets should be converted to a plain text format, such as comma-separated values (.csv) or tab-delimited ASCII text (.txt).
3. Software files should be submitted both as compiled code and as source code. Submitting compiled code without accompanying source code is not acceptable.
4. File names should follow these conventions:
    a. the allowable characters are the letters ("a-z" "A-Z"), the numerals ("0-9"), the hyphen ("-"), the underscore ("_"), and the period (".")
    b. the length should not exceed 35 characters
    c. dates should follow the YYYY-MM-DD number format
    d. words should be separated by the hyphen ("-")
    e. categorical groupings of files should make use of the underscore ("_")

    For example:

| | |
|---|---|
| siteA_2020-04-01_animal-count.csv | siteA_2020-05-01_animal-count.csv |
| siteB_2020-04-01_animal-count.csv | siteB_2020-04-01_weather.dat |
| siteB_2020-05-01_animal-count.csv | siteB_2020-05-01_weather.dat |